29/7/26

Stack queue and Linked List

① Push(52)', Push(52)', PoP(); Push(55); Push(62)

S = PoP();

Stack operation:

Push(54) → [54]

Push(52) → [54, 52]

Push() → remove 52

Push(55) → [54, 55]

Push(62) → [54, 55, 62]

S = PoP() → S = 62

Queue operation:

- enqueue(21) → [21]

enqueue(24) → [21, 24]

dequeue() → remove 21

enqueue(28) → [24, 28]

enqueue(32) → [24, 28, 32]

q = dequeue() → q = 24

Ans

S + q = 62 + 24 = 86

②

10 5 + 60 6 / * 8 −

$\frac{5}{10}$ + ⇒ 15 ; $\frac{6}{60}$ ⇒ 10

$\frac{15}{10}$ × = 150;

$\frac{8}{150}$ − ⇒ 142.

③ 6 enqueue operations,

④ It is a true Statement

* Both Push and PoP operations can be implemented in O(1) time. Reverse can also be implemented in O(1) time by maintaing the stack orientation

⑤ 8 2 3 ^ / 2 3 * + 5 1 * −

Push 8, 2, 3 → ^ : $2^3 = 8$

⇒ stack [8, 8]

8/8 = 1 → [1]

Push 2, 3 → [1, 2, 3]

→ first * : 2×3 = 6

stack becomes [1, 6]

$\boxed{top = 6}$, $\boxed{next = 1}$

⑥ infix:

a + b × c − d ^ e ^ f

since ^ is right associative:

d^(e^f)

Postfix = abc * + def^^ _ *

⑦ Push (10)

Push (20)

Pop → 20

Push (10)

Push (20)

Pop → (20)

Pop → 10

Pop → 10

Push → (20)

Pop → 20.

Sequence is 20, 20, 10, 10, 20.